

Find My Polynomial

Fall 2024 ARML Power Contest

The Solutions

1. $A = p(1) = a_k \cdot 1^k = a_k$. Then $B = a_k \cdot 2^k = A \cdot 2^k$, so $2^k = B/A$. Thus $k = \log_2(B/A)$ and $a_k = A$.
2. Once Bob learns that $p(x) = a_k x^k = A$, he knows that $a_k = A/x^k$. Unfortunately, he doesn't know the value of k , so he can't be sure what the monomial is. For example, let's say that Bob picks $x = 3$ and learns that $p(3) = 36$. Two possibilities for $p(x)$ are $12x$ and $4x^2$. If Bob chooses $x = c$ and thinks the monomial might be $a_k x^k$ it could just as well be $(ca_k)x^{k-1}$ or $(a_k/c)x^{k+1}$.
3. A polynomial of degree at most 1 is of the form $a_1x + a_0$, a linear polynomial. Since two points determine a line Bob expects to be able to figure out the polynomial with two pieces of information. In fact, $p(0) = A$, so $a_0 = A$. Then $p(1) = B$. Since $p(1) = a_1 + a_0 = a_1 + A$, Bob can easily compute that $a_1 = B - A$. Thus, Alice's polynomial is $(B - A)x + A$.
4. Similarly, a polynomial of degree 2 or less (a quadratic polynomial, if degree is 2) is of the form $a_2x^2 + a_1x + a_0$. As before, $p(0) = a_0 = A$. This time, $p(1) = a_2 + a_1 + A = B$ and $p(-1) = a_2 - a_1 + A = C$. Subtracting the second from the first yields $2a_1 = B - C$, so $a_1 = \frac{B-C}{2}$. Adding the two equations gives $2a_2 + 2A = B + C$ so $a_2 = \frac{B+C-2A}{2}$.
5. Substituting $x = r_i$ with $i \neq j$ causes one of the terms in the numerator of $\ell_j(x)$ to be zero, so $\ell_j(r_i) = 0$.
6. Substituting $x = r_j$ causes each term in the numerator to equal the term below it in the denominator. Since, as noted in the problem text, no term is zero, all the factors can be canceled and $\ell_j(r_j) = 1$.
7. Bob can choose any $k + 1$ distinct numbers $r_0, \dots, r_k$. (For instance, $r_0 = 0, r_1 = 1, \dots, r_k = k$ is a good choice.) Let $A_j = p(r_j)$. Then let $L(x) = A_0\ell_0(x) + A_1\ell_1(x) + \dots + A_k\ell_k(x)$. Claim: $L(x) = p(x)$. First, note that $L(x)$ is a polynomial of degree k or less, being a sum of constant multiples of polynomials of degree k . Second, $L(r_i) = A_0\ell_0(r_i) + \dots + A_k\ell_k(r_i)$. By the previous two problems, $\ell_j(r_i) = 0$ if $i \neq j$ but $\ell_i(r_i) = 1$. So $L(r_i) = A_i = p(r_i)$. So $L(x)$ and $p(x)$ agree at $k + 1$ different values of x . Their difference is a polynomial of degree at most k which is zero at these $k + 1$ values of x , so the difference is zero and $L(x) = p(x)$.
8. Assume Bob asks for the k values $p(r_1), p(r_2), \dots, p(r_k)$. Choose any r_0 distinct from $r_1, r_2, \dots, r_k$. Then adding any multiple of $\ell_0(x)$ to $p(x)$ creates another polynomial of degree at most k with the exact same values as $p(x)$ at $r_1, \dots, r_k$. So there are many different polynomials whose values agree with $p(x)$ for the questions that Bob asks, so Bob cannot determine which polynomial Alice has chosen from just the k questions he asked.

9. The polynomial is $3x^8 + 6x^7 + 4x^6 + 2x^5 + 8x^3 + 5x^2 + 9x + 2$.

Note that $p(x) = a_k x^k + a_{k-1} x^{k-1} + \cdots + a_1 x + a_0 = x(a_k x^{k-1} + a_{k-1} x^{k-2} + \cdots + a_1) + a_0$. Substituting $x = 10$ makes $p(10) = 10A + a_0$ where $A = a_k \cdot 10^{k-1} + a_{k-1} \cdot 10^{k-2} + \cdots + a_1$ is a nonnegative integer. Thus $10A + a_0 = 364,208,592 = 10 \cdot 36,420,59 + 2$. This can be rewritten as $10(A - 36,420,59) = 2 - a_0$. The left-hand side is a multiple of 10 while the right-hand side is some integer between -7 and 2 inclusive (because a_0 is a non-negative integer less than 10). The only way this equation could be true is if both sides are zero, requiring $a_0 = 2$ and $A = 36,420,59$. Now repeat this process on A to show that $a_1 = 9$, and so forth.

Basically, this is the process for converting a number into base 10.

10. Let $p(x) = a_k x^k + a_{k-1} x^{k-1} + \cdots + a_1 x + a_0$. Note that $p(2) = a_k \cdot 2^k + \cdots + a_1 \cdot 2 + a_0 = 2(a_k \cdot 2^{k-1} + \cdots + a_1) + a_0 = 2A + a_0 = 593$, where A represents the term in the parentheses, which is a nonnegative integer. Since $2A$ is even and a_0 is either 0 or 1, and 593 is odd, a_0 must be 1. Now subtract 1 from both sides of this equation and divide by 2 to obtain $a_k \cdot 2^{k-1} + a_{k-1} \cdot 2^{k-2} + \cdots + a_2 \cdot 2 + a_1 = 296$.

The terms excluding a_1 on the left-hand side can have 2 factored out: $2(a_k \cdot 2^{k-2} + \cdots + a_2) + a_1 = 296$. Once again, the left-hand side is an even number plus a_1 , while the right-hand side is even. Since a_1 is either 0 or 1, it must be the case that $a_1 = 0$.

This process of dividing by 2 and isolating one coefficient continues—basically this is the process for converting a number into binary notation. Since $593 = 1001010001_2$, Alice's polynomial must be $x^9 + x^6 + x^4 + 1$.

11. As noted in the previous problem, the process for finding the coefficients by repeatedly dividing by $x = 2$ is essentially converting the value of $p(2)$ into binary notation. So Bob's strategy is: write $p(2)$ in binary and take the digits as the coefficients in Alice's polynomial, with increasing significance of the digit corresponding to increasing powers of x in the polynomial.
12. This time, Bob should ask for $p(b)$ and convert the result to base- b notation. As in the previous problem, the base- b digits become the coefficients of the polynomial. To be specific, let $p(b) = N$ and write $N = qb + r$ where q and r are the quotient and remainder on division of N by b . Then

$$\begin{aligned} p(b) &= a_k b^k + a_{k-1} b^{k-1} + \cdots + a_1 b + a_0 = qb + r = N \\ b(a_k b^{k-1} + a_{k-1} b^{k-2} + \cdots + a_1) + a_0 &= qb + r \\ b(a_k b^{k-1} + \cdots + a_1 - q) &= r - a_0 \end{aligned}$$

The left-hand side of the last equation is divisible by b . Since both r and a_0 are numbers between 0 and $b-1$ (inclusive) the only way their difference could be divisible by b is if it is zero. So $a_0 = r$. Now divide both sides by b , leaving $a_k b^{k-1} + \cdots + a_2 b + a_1 = q$ and repeat the process. Since $0 \leq q < N$ the number on the right-hand side gets smaller each time through the process, so eventually the quotient is zero and all further base- b digits (and all further coefficients) must be zero.

13. Bob first asks Alice for $p(1)$. Note that $p(1) = a_k + a_{k-1} + \cdots + a_1 + a_0$ is the sum of the coefficients of her polynomial. Since each coefficient is nonnegative, $b = p(1) + 1$ is a positive integer greater than any of the coefficients. Thus, Bob need only use his second question to ask for $p(b)$ and use the method of the previous problem to find Alice's polynomial.
14. The process is similar to the previous few problems, only this time Bob needs to use remainders of -1 , 0 , and 1 (instead of 0 , 1 , and 2) when he divides by 3 . Thus

$$\begin{aligned} p(3) &= a_k \cdot 3^k + a_{k-1} \cdot 3^{k-1} + \cdots + a_1 \cdot 3 + a_0 = 1573 = 3 \cdot 524 + 1 \\ 3(a_k \cdot 3^{k-1} + \cdots + a_1) + a_0 &= 3 \cdot 524 + 1 \end{aligned}$$

Since a_0 is -1 , 0 , or 1 , the only possible choice is $a_0 = 1$ and then 1 can be canceled from both sides of the equation, and then both sides divided by 3 to obtain

$$\begin{aligned} a_k \cdot 3^{k-1} + a_{k-1} \cdot 3^{k-2} + \cdots + a_2 \cdot 3 + a_1 &= 524 \\ 3(a_k \cdot 3^{k-2} + a_{k-1} \cdot 3^{k-3} + \cdots + a_3 \cdot 3 + a_2) + a_1 &= 3 \cdot 175 - 1 \end{aligned}$$

Note that this time, when 524 was divided by 3 , instead of writing it as $524 = 3 \cdot 174 + 2$ the division was computed as $3 \cdot 175 - 1$. This is because the remainders are not allowed to be 2 , while -1 is allowed instead. Thus, any time a remainder of 2 would appear, the quotient should be increased by 1 while using a remainder of -1 instead.

So Bob learns that $a_1 = -1$, and then $a_k \cdot 3^{k-2} + \cdots + a_3 \cdot 3 + a_2 = 175$. Bob continues to divide by 3 ; if the remainder is 0 or 1 , he sets the next coefficient equal to that remainder and divides the quotient by 3 to continue. If the remainder is 2 , Bob increases the quotient by 1 so that the remainder becomes -1 and then divides the new quotient by 3 to continue. Thus, Bob's calculations are:

$$\begin{array}{ll} 1573 = 3 \cdot 524 + 1 & \rightarrow a_0 = 1 \\ 524 = 3 \cdot 175 - 1 & \rightarrow a_1 = -1 \\ 175 = 3 \cdot 58 + 1 & \rightarrow a_2 = 1 \\ 58 = 3 \cdot 19 + 1 & \rightarrow a_3 = 1 \\ 19 = 3 \cdot 6 + 1 & \rightarrow a_4 = 1 \\ 6 = 3 \cdot 2 + 0 & \rightarrow a_5 = 0 \\ 2 = 3 \cdot 1 - 1 & \rightarrow a_6 = -1 \\ 1 = 3 \cdot 0 + 1 & \rightarrow a_7 = 1 \end{array}$$

Since all further divisions end with both quotient and remainder zero, there are no further coefficients. Thus Alice's polynomial is $x^7 - x^6 + x^4 + x^3 + x^2 - x + 1$.

The notation of expressing numbers in base-3 using the digits -1 , 0 , and 1 is called *balanced ternary* notation.

15. The method continues to remain the same. Since there are five numbers in a row (-1 , 0 , 1 , 2 , and 3) that the coefficients are allowed to be, use a method similar to the

foregoing problems, dividing by 5. Any remainder of 4 will require that the quotient increase by 1 so that the remainder can be -1 instead. So

$$\begin{array}{ll} 1238 = 5 \cdot 247 + 3 & \rightarrow a_0 = 3 \\ 247 = 5 \cdot 49 + 2 & \rightarrow a_1 = 2 \\ 49 = 5 \cdot 9 + 4 = 5 \cdot 10 - 1 & \rightarrow a_2 = -1 \\ 10 = 5 \cdot 2 + 0 & \rightarrow a_3 = 0 \\ 2 = 5 \cdot 0 + 2 & \rightarrow a_4 = 2 \end{array}$$

and all quotients and remainders from this point on are zero. So Alice's polynomial is $2x^4 - x^2 + 2x + 3$.

16. Since the interval does not contain the numbers m or n , the integers in the range are all between $\lceil m \rceil$ and $\lfloor n \rfloor$ inclusive. Let $b = \lfloor n \rfloor - \lceil m \rceil + 1$ and proceed by dividing by b . If the remainder is not in the interval (m, n) then add or subtract b from the remainder until it is in this range, respectively subtracting or adding 1 to the quotient each time. There are exactly b integers in the interval (m, n) so it is guaranteed (by the pigeonhole principle) that the remainder plus or minus some multiple of b will be an integer in the given range. Once this number is found, that will become the value of the next coefficient in Alice's polynomial, and Bob then continues by dividing the quotient (as altered) by b again. The process repeats until the quotient becomes zero.
17. One solution method is as follows. First, since no coefficient of $p(x)$ is smaller than $-n$, the polynomial $p^*(x) = p(x + n + 1)$ has all nonnegative coefficients (and the same positive leading coefficient). This fact will be proven below. Now the method of problem 13 will determine $p^*(x)$ with just two questions, and then all Bob needs to do is to compute $p(x) = p^*(x - n - 1)$.

Proof that $p^*(x) = p(x + n + 1)$ has all coefficients nonnegative.

Suppose $p(x) = a_k x^k + a_{k-1} x^{k-1} + \cdots + a_1 x + a_0$, with a_k a positive integer and $a_j \geq -n$ for $j < k$. The statement will be proven by induction on k .

For $k = 1$, the polynomial $p^*(x) = p(x + n + 1) = a_1(x + n + 1) + a_0 = a_1 x + a_1(n + 1) + a_0$. Since $a_1 \geq 1$ and $a_0 \geq -n$, the constant term is at least $n + 1 - n > 0$.

Now assume that for any polynomial $p(x)$ of degree k satisfying the hypotheses that $a_k > 0$ and $a_j \geq -n$ for all $j < k$, all of the coefficients of $p^*(x) = p(x + n + 1)$ are positive. Then for a degree $k + 1$ polynomial,

$$\begin{aligned} p^*(x) &= p(x + n + 1) = a_{k+1}(x + n + 1)^{k+1} + a_k(x + n + 1)^k + \cdots + a_1(x + n + 1) + a_0 \\ &= a_{k+1}(x + n + 1)^{k+1} - (n + 1)(x + n + k)^k \\ &\quad + (n + 1 + a_k)(x + n + 1)^k + a_{k-1}(x + n + 1)^{k-1} + \cdots + a_0 \end{aligned}$$

The first line in the final right-hand side of this equation can be rewritten as

$$(a_{k+1}(x + n + 1) - (n + 1))(x + n + k)^k.$$

Since a_{k+1} is assumed to be a positive integer, all terms in this expression are positive. And since $a_k > -n$, the expression in the bottom line of the right-hand side satisfies the conditions of the induction hypothesis, so all of its coefficients are positive as well. Adding these two results in a polynomial with all positive coefficients, so the induction is complete.